Metric audit and evaluation rationale

Audit outcome

Both saved checkpoints were evaluated again on the exact released labels, with no retraining or threshold optimization on the test set. Original ROC AUC and AP values reproduce: maximum per-complex absolute errors are 0 and

0. AUC and AP were also recomputed from independent mathematical formulas, including tied scores; maximum discrepancy from scikit-learn is 3.33e-16.

The headline ranking numbers were numerically correct. The old evaluation was incomplete for judging whether the highest-ranked contacts are useful. It also needed explicit tie handling, score-calibration warnings, and a clearer distinction between AP and trapezoidal PR area.

The practical conclusion must include retrieval: GCN top-10 precision is 0.04909, versus 0.05091 for the MLP: 27.0 versus 28.0 expected true contacts among 550 recommendations (ten per complex). The GCN finds no contact in the top ten on 39 complexes, compared with 36 for the MLP. GCN top-10 precision is lower; its paired interval is [-0.02727272727272726, 0.021818181818181827]. The GCN's partial ROC AUC at low FPR is lower. These measures must be considered alongside the overall ROC AUC, rather than declaring a uniform improvement from the headline score alone.

What was checked

- Both compressed datasets match the authors' published MD5 hashes.
- Raw labels contain exactly -1 and +1; +1 is the contact class. Pair indices are within the corresponding ligand/receptor feature arrays. No labeled pair repeats inside a complex.
- The fixed 140/35 development split and the 55 official test IDs match the saved experiment. Development and test IDs are disjoint.
- The test includes 4,958,317 released pairs and 4,871 contacts. Unlisted Cartesian-product pairs are excluded, rather than silently converted to negatives.
- Exact identical feature-and-neighborhood representations across development and test: 0. This is an exact-array check, not a sequence-homology audit.
- The biological 6 Å contact definition comes from the data authors. This audit verifies the released labels and their use, not a fresh reconstruction of bound/unbound residue correspondences from experimental structures.
- Saved model/data SHA-256 hashes, package versions, and pair-level outputs make the recomputation traceable in results/metric_audit.json and results/audit_predictions.

Why these metrics are included

Metric	Question answered	Limitation
Median complex ROC AUC	Does a contact usually rank above a noncontact within a typical complex?	Paper-comparable ranking; says little about precision at the top.
AP	How well are contacts concentrated near the top across the full ranking?	Depends on contact prevalence; compare models on identical labels.
Trapezoidal PR area	What is the area under the linearly joined PR points?	A different numerical convention from AP; never call them interchangeable.
Precision/recall at 10, 50, L/5, L	If we inspect a limited number of pairs per complex, how many contacts do we recover?	K is a stated budget; L=min(partner residue counts), L/5=floor(L/5), at least 1.
Top-K enrichment	How much better is top-K precision than random selection in that complex?	Relative lift can look large while absolute precision is low.
Partial ROC AUC at FPR <=1%	Does ranking remain useful when false-positive rates must be small?	Standardized scikit-learn convention; chance=0.5.
Confusion counts, precision, recall, F1, MCC	What happens at a stated score threshold?	Here 0.5 is diagnostic only; no deployment threshold has been validated.
Brier score and log loss	Can the weighted sigmoid outputs safely be interpreted as probabilities?	Diagnostic only; outputs were not calibrated on a naturally imbalanced holdout.

Recomputed ranking results

Ranking uses raw logits to avoid extra rounding/saturation ties from sigmoid. The old sigmoid-based metrics are retained separately for reproduction. The added retrieval budgets and diagnostic metrics were specified for this post-hoc audit; they are exploratory, not preregistered confirmatory endpoints.

Measure	MLP	GCN
Median per-complex ROC AUC	0.843190	0.891311
Median per-complex AP	0.012737	0.018246
Median trapezoidal PR area	0.011289	0.017445
Mean precision at 10	0.050909	0.049091
Mean recall at 10	0.006787	0.006579
Mean precision at L/5	0.044822	0.050522
Mean precision at L	0.033723	0.036760
Pooled ROC AUC	0.844794	0.873831
Pooled AP	0.008346	0.009276
Pooled trapezoidal PR area	0.008321	0.009239
Standardized pooled partial ROC AUC (FPR <= 1%)	0.542150	0.541595

Top-K precision uses the expected number of contacts under uniform selection inside a tie at the cutoff. This prevents positive-first input ordering from deciding which tied predictions receive credit. Each complex contributes equally to macro means/medians. Pooled metrics rank all test pairs together; large complexes and score offsets

between complexes can change them.

Fixed-score classification diagnostic

The following uses sigmoid score >=0.5. It is not an independently validated decision rule and must not be presented as optimal operating performance.

Model	TP	FP	FN	TN	Precision	Recall	F1	MCC
MLP	3,923	1,437,314	948	3,516,132	0.002722	0.805379	0.005426	0.035546
GCN	2,702	475,316	2,169	4,478,130	0.005653	0.554712	0.011191	0.048693

An always-negative classifier has test accuracy 99.901761%, recall 0, F1 0 and MCC 0. This is why plain accuracy is not the headline metric. Sensitivity, specificity, balanced accuracy and the confusion counts are retained in the JSON/CSV.

Probability interpretation

Development pairs were downsampled to about 10 negatives per contact, and the loss gives positives weight 10. That combination changes the effective training prior. A sigmoid score of 0.8 is therefore not evidence of an 80% contact probability under the test prevalence. The audit reports Brier/log-loss diagnostics without fitting a calibrator on test labels. A useful threshold and calibration require independent validation data with the natural pair distribution; the released development files do not provide that distribution directly.

Score source	Mean score	Brier score (lower is better)	Log loss (lower is better)
MLP raw sigmoid	0.319767	0.176810	0.511639
GCN raw sigmoid	0.127628	0.066811	0.214964
Empirical-prevalence constant (diagnostic)	0.000982	0.000981	0.007785

The constant reference uses the observed test prevalence after the fact. It illustrates the scale of probability error; it is not a separately trained deployable baseline. The raw neural scores should be used for ranking until calibration is evaluated on suitable independent validation data.

Uncertainty and remaining limits

Paired bootstrap intervals resample complexes, not millions of dependent residue pairs. Full-set and the fixed 31-case historical subset are in the JSON. These intervals describe variation across the observed complexes conditional on the two trained models. They do not measure training-seed variation or establish sequence-family independence. All 55 cases have now been examined; the 31-case historical subset is not a new blind test. One training seed and prior test exposure remain study limitations even after the metric code is correct.

The MLP has 21,121 trainable parameters; the GCN has 29,697. They use matched feature/head widths, not matched parameter counts. Thus this comparison supports an association with graph context but does not isolate it from added capacity.

Reproduce this audit

Run python paper_benchmark/test_evaluation.py and python paper_benchmark/audit_metrics.py. The second command checks cached prediction provenance before reuse; delete that cache to force fresh inference. The original experiment JSON and checkpoints are preserved.

Sources: paper metric definition, AP definition, partial ROC AUC, probability calibration.